INCH-POUND

MIL-DTL-3933/10F
19 November 2009
SUPERSEDING
MIL-DTL-3933/10E
25 July 2006

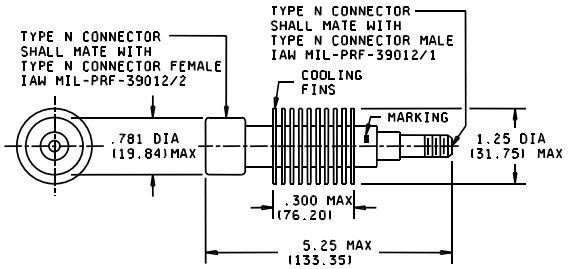
DETAIL SPECIFICATION SHEET

ATTENUATORS, FIXED COAXIAL LINE (SERIES N),

FREQUENCY RANGE: DC TO 18 GHz, CLASS III, MEDIUM POWER

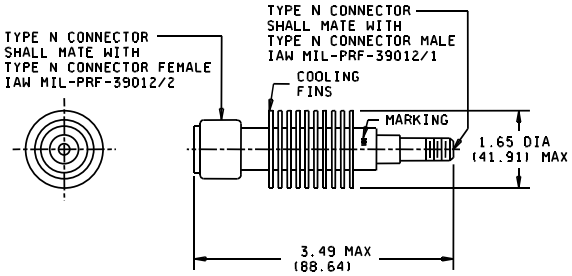
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration for PIN M3933/10-1.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 2. Dimensions and configuration for PIN M3933/10-2 through M3933/10-22.

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ENGINEERING DATA:

Attenuators for use in general field equipment.

Operating frequency range: DC to 18 gigahertz (GHz) (see table I).

Test frequency range for VSWR and attenuation measurements, measured or recorded, when using an Automatic Network Analyzer (ANA) or equivalent test equipment:

100 MHz (or lower) to	Dash numbers
4 GHz	01 through 05
8 GHz	06 through 10
11 GHz	11 through 15
18 GHz	16 through 22

Basic test frequency for sensitive measurements like connector repeatability, temperature sensitivity, power sensitivity, and any other measurements where a 'delta' specification is imposed:

Basic test frequency	Dash numbers
2 GHz	01 through 05
4 GHz	06 through 10
6 GHz	11 through 15
8 GHz	16 through 22

Operating temperature range:

M3933/10-1: -55°C to +85°C.

M3933/10-2 through -22: -55°C to +125°C.

Maximum weight: 6 ounces.

REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2. Cooling fins shall be circular.

Nominal impedance: 50 ohms.

Attenuation: See table I. Stability and sensitivity:

Maximum attenuation change after:	Up to 10 dB, inclusive (dB)	Over 10 dB (dB/dB)
Temperature change/ thermal shock	0.05	0.005
Vibration or shock	0.1	0.01
Moisture resistance or salt spray	0.2	0.02
Peak power	0.05	0.005

Frequency sensitivity, maximum: 0.1 dB/dB/GHz.
Temperature sensitivity of attenuation, maximum: 0.0004 dB/dB/°C.

Power: See table I.
Power sensitivity for full input power, maximum: 0.005 dB/dB/Watt.

VSWR: See table I.

Connector repeatability, maximum variation in attenuation: 0.02 dB.
Part or Identifying Number (PIN): M3933/10- (and dash number from table I).

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TABLE I. Electrical characteristics.

Dash number	Attenuation (dB)							Maximum power Input		VSWR maximum 1/					
	Nominal	Deviation													
		DC to 4.0 GHz	DC to 6 GHz	DC to 8 GHz	DC to 11 GHz	DC to 12 GHz	DC to 18.0 GHz	AV (W) at 25°C (continuously)	Peak 2/ (kW)	DC to 4.0 GHz	4.0 to 6 GHz	4.0 to 8.0 GHz	6.0 to 11 GHz	8.0 to 12.0 GHz	12.0 to 18.0 GHz
01 N S	6	±0.2	---	---	---	---	---	15	3 2/	1.30:1	---	---	---	---	---
02 N S	3	±0.5	---	---	---	---	---	20 4/	1 3/	1.15:1	---	---	---	---	---
03 N S	10	±0.5	---	---	---	---	---	20 4/	1 3/	1.15:1	---	---	---	---	---
04 N S	20	±0.5	---	---	---	---	---	20 4/	1 3/	1.15:1	---	---	---	---	---
05 N S	30	±0.75	---	---	---	---	---	20 4/	1 3/	1.15:1	---	---	---	---	---
06 N S	3	±0.3	---	±0.6	---	---	---	25 5/	5 2/	1.20:1	---	1.30:1	---	---	---
07 N S	6	±0.3	---	±0.6	---	---	---	25 5/	5 2/	1.20:1	---	1.30:1	---	---	---
08 N S	10	±0.3	---	±0.6	---	---	---	25 5/	5 2/	1.20:1	---	1.30:1	---	---	---
09 N S	20	±0.3	---	±0.6	---	---	---	25 5/	5 2/	1.20:1	---	1.30:1	---	---	---
10 N S	30	±0.6	---	±1.0	---	---	---	25 5/	5 2/	1.20:1	---	1.30:1	---	---	---
11 N S	3	±0.35	±0.65	---	±0.75	---	---	20 4/	1 2/	1.15:1	1.20:1	---	1.30:1	---	---
12 N S	6	±0.35	±0.65	---	±0.75	---	---	20 4/	1 2/	1.15:1	1.20:1	---	1.30:1	---	---
13 N S	10	±0.35	±0.65	---	±0.75	---	---	20 4/	1 2/	1.15:1	1.20:1	---	1.30:1	---	---
14 N S	20	±0.35	±0.65	---	±0.75	---	---	20 4/	1 2/	1.15:1	1.20:1	---	1.30:1	---	---
15 N S	30	±0.35	±0.65	---	±0.75	---	---	20 4/	1 2/	1.15:1	1.20:1	---	1.35:1	---	---

See footnotes at end of table I.

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation (dB)							Maximum power Input		VSWR maximum 1/					
	Nominal	Deviation													
		DC to 4.0 GHz	DC to 6 GHz	DC to 8 GHz	DC to 11 GHz	DC to 12 GHz	DC to 18.0 GHz	AV (W) at 25°C (continuously)	Peak 2/ (kW)	DC to 4.0 GHz	4.0 to 6 GHz	4.0 to 8.0 GHz	6.0 to 11 GHz	8.0 to 12.0 GHz	12.0 to 18.0 GHz
16 N S	3	---	---	---	---	---	±0.3	20 5/	1 2/	1.15:1	---	1.20:1	---	1.25:1	1.40:1
17 N S	6	---	---	---	---	---	±0.3	10 5/	1 2/	1.15:1	---	1.20:1	---	1.25:1	1.40:1
18 N S	10	---	---	---	---	---	±0.5	10 5/	2 2/	1.15:1	---	1.20:1	---	1.25:1	1.40:1
19 N S	20	---	---	---	---	---	±0.5	10 5/	2 2/	1.15:1	---	1.20:1	---	1.25:1	1.40:1
20 N S	30	---	---	---	---	---	±1.0	10 5/	2 2/	1.15:1	---	1.20:1	---	1.25:1	1.40:1
21 N S	40	---	---	---	---	---	±1.0	10 5/	2 2/	1.15:1	---	1.20:1	---	1.25:1	1.40:1
22 N S	50	---	---	---	---	---	±1.35	10 5/	2 2/	1.15:1	---	1.20:1	---	1.25:1	1.40:1

1/ VSWR value is for both ends.

2/ Peak power for a duty cycle of 5×10^{-4} maximum duration 5 microseconds.

3/ Peak power for a duty cycle of 1×10^{-3}.

4/ Derate linearly from 20 watts at 25°C to 10 percent at 125°C.

5/ Derate linearly from the stated voltage at 25°C to 0 watts at 125°C.

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Referenced documents. In addition to MIL-DTL-3933, this specification sheet references the following documents:
MIL-PRF-39012/1
MIL-PRF-39012/2

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5985-2009-033)

Review activities:
Army - MI
Navy - AS, MC, SH
Air Force - 99

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